

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 21

In this class we prove the fundamental theorem of Galois theory, and introduce what it means to solve a polynomial equation in radicals.

1.1. The fundamental theorem of Galois theory. We recall the statement of the fundamental theorem of Galois theory from last class. If K is a number field, E/K is a finite normal extension of K , $H \leq \text{Gal}(E/K)$ is a subgroup of the Galois group, and $K \leq L \leq E$ is a subfield of E that contains K then we define:

$$(1.1) \quad E^H = \{\alpha \in E \mid h(\alpha) = \alpha, \text{ for all } h \in H\},$$

$$(1.2) \quad H_L = \{h \in \text{Gal}(E/K) \mid h(\ell) = \ell, \text{ for all } \ell \in L\}.$$

As we have seen in the homework, E^H is a subfield of E and H_L is a subgroup of $\text{Gal}(E/K)$. The main point of the fundamental theorem of Galois theory is that these operations are inverses: $H_{E^H} = H$, and $E^{H_L} = L$ for subgroups $H \leq \text{Gal}(E/K)$ and subfields $K \leq L \leq E$.

Theorem 1.1 (Fundamental theorem of Galois theory). *Let K be a number field and let E/K be a finite normal extension of K . The following holds:*

- (i) *If $K \leq L \leq E$ is a subfield then there is a subgroup $H \leq \text{Gal}(E/K)$ such that $L = E^H$.*
- (ii) *If $H \leq \text{Gal}(E/K)$ then there is a subfield $K \leq L \leq E$ such that $H = H_L$.*
- (iii) *The correspondence between subfields and subgroups are inverses: If $H \leq \text{Gal}(E/K)$ then $H_{E^H} = H$, and if $K \leq L \leq E$ is a subfield then $E^{H_L} = L$.*
- (iv) *A subgroup $H \leq \text{Gal}(E/K)$ is a normal subgroup if and only if $K \leq E^H \leq E$ is a finite normal extension of K .*
- (v) *If $K \leq L \leq E$ is a finite normal extension of K then*

$$\text{Gal}(L/K) \cong \text{Gal}(E/K)/H_L,$$

where the quotient $\text{Gal}(E/K)/H_L$ is a group since $H_L \leq \text{Gal}(E/K)$ is a normal subgroup.

We will prove this theorem in steps, we begin by two lemmas which will be very important when we prove part (iii) (as well as (i) and (ii)).

Lemma 1.1. *Let K be a number field and E/K a finite normal extension of K . If $K \leq L \leq E$ is a subfield of E , and if $\alpha, \beta \in E$ are conjugates over L , then there exists some $\phi \in H_L$ such that $\phi(\alpha) = \beta$.*

Proof. Consider the isomorphism $\psi_{\alpha, \beta} : L(\alpha) \rightarrow L(\beta)$. By the extension theorem we can extend $\psi_{\alpha, \beta}$ to some isomorphism $\phi : E \rightarrow E'$ where $L(\beta) \leq E'$ and $\phi|_{L(\alpha)} = \psi_{\alpha, \beta}$. We claim that $E' = E$. Indeed, for $a \in K$ we have $\phi(a) = \psi_{\alpha, \beta}(a) = a$ since $\psi_{\alpha, \beta}$ fix L , and $K \leq L$. So, for any $\gamma \in E$ let $f_\gamma(x) = a_0 + \dots + a_n x^n$, where $a_j \in K$ so $\phi(a_j) = a_j$, be the irreducible polynomial for γ over K , then:

$$\begin{aligned} f_\gamma(\phi(\gamma)) &= a_0 + \dots + a_n \phi(\gamma)^n = \phi(a_0) + \dots + \phi(a_n) \phi(\gamma)^n = \\ &= \phi(a_0 + \dots + a_n \gamma^n) = \phi(f_\gamma(\gamma)) = \phi(0) = 0. \end{aligned}$$

It follows that $\phi(\gamma)$ is a zero of $f_\gamma(x)$, so since E is a finite normal extension of K , and since $f_\gamma(x)$ is irreducible over K , it follows that $\phi(\gamma) \in E$. That is, the image of ϕ is contained in E , so since ϕ is an isomorphism we see that $\phi : E \rightarrow E$. Finally, note that for any $\eta \in L$ we have $\phi(\eta) = \psi_{\alpha, \beta}(\eta) = \eta$, and $\phi(\alpha) = \psi_{\alpha, \beta}(\alpha) = \beta$ so $\phi \in H_L$ and ϕ maps α to β . ■

Lemma 1.2. *Let K be a number field and E/K a finite normal extension of K . If $H \leq \text{Gal}(E/K)$ is a subgroup and $\alpha \in E$ then the polynomial:*

$$(1.3) \quad f_{H, \alpha}(x) = \prod_{h \in H} (x - h(\alpha))$$

is a polynomial with coefficients in E^H , $f_{H, \alpha}(x) \in E^H[x]$.

Proof. To simplify notation we define for any $h \in H$ the map $h : E[x] \rightarrow E[x]$, $h(a_0 + a_1 x + \dots + a_m x^m) = h(a_0) + h(a_1)x + \dots + h(a_m)x^m$. Recall from the homework that $h : E[x] \rightarrow E[x]$ is an isomorphism of rings for every $h \in H$ since $h : E \rightarrow E$ is an automorphism (and therefore an isomorphism of fields). For any $h \in H$ we have:

$$\begin{aligned} h(f_{H, \alpha}(x)) &= h\left(\prod_{h' \in H} (x - h'(\alpha))\right) = \prod_{h' \in H} h(x - h'(\alpha)) = \\ &= \prod_{h' \in H} (x - (hh')(\alpha)) = \prod_{h'' \in H} (x - h''(\alpha)) = f_{H, \alpha}(x). \end{aligned}$$

It follows that if we write $f_{H, \alpha}(x) = a_0 + a_1 x + \dots + a_m x^m$, $m = |H|$, then for any $h \in H$:

$$a_0 + a_1 x + \dots + a_m x^m = f_{H, \alpha}(x) = h(f_{H, \alpha}(x)) = h(a_0) + h(a_1)x + \dots + h(a_m)x^m$$

or

$$(a_0 - h(a_0)) + (a_1 - h(a_1))x + \dots + (a_m - h(a_m))x^m = 0$$

which implies that $h(a_j) = a_j$ for each $j = 1, \dots, m$. Since this holds for any $h \in H$ we conclude that we have $h(a_j) = a_j$ for every $h \in H$ which implies that $a_j \in E^H$. Since all the coefficients of $f_{H, \alpha}(x)$ are in E^H we conclude that $f_{H, \alpha}(x) \in E^H[x]$ and the lemma follows. ■

We can now prove Theorem 1.1.

Proof of Theorem 1.1. We begin by proving that $E^{H_L} = L$ for any subfield $K \leq L \leq E$. Notice that any element $h \in H_L$ by definition fix L , so we have $L \leq E^{H_L}$ and only need to prove the converse inclusion. Assume for contradiction that we have $\alpha \in E^{H_L} \setminus L$. Let β be a conjugate of α over L such that $\alpha \neq \beta$ (such β must exist since otherwise we would have $\alpha \in L$). By Lemma 1.1 there is some $\phi \in H_L$ such that $\phi(\alpha) = \beta$. Since $\beta \neq \alpha$ this implies that there is some $\phi \in H_L$ such that $\phi(\alpha) \neq \alpha$ which implies that $\alpha \notin E^{H_L}$. This is a contradiction, and we conclude that $E^{H_L} = L$.

If $H \leq \text{Gal}(E/K)$, let E^H be the corresponding field. Since E is a finite extension of E^H the primitive element theorem implies that there is some $\theta \in E$ such that $E = E^H(\theta)$. Now consider the polynomial $f_{H,\theta}(x) \in E^H[x]$ from Lemma 1.2. If $\varphi \in H_{E^H}$ is arbitrary then, since $f_{H,\theta}(x)$ has coefficients in E^H and φ fix E^H , we have:

$$\prod_{h \in H} (x - (\varphi \circ h)(\theta)) = \varphi \left(\prod_{h \in H} (x - h(\theta)) \right) = \varphi(f_{H,\theta}(x)) = f_{H,\theta}(x) = \prod_{h \in H} (x - h(\theta))$$

which implies, in particular, that the factor $x - \varphi(\theta)$ on the left must occur as some factor $x - h(\theta)$ on the right. That is, there is some $h \in H$ such that $h(\theta) = \varphi(\theta)$. Now, notice that the map $h^{-1} \circ \varphi$ fix L (since both h and φ fix L), and fix θ since $h(\theta) = \varphi(\theta)$. It follows that $h^{-1} \circ \varphi$ fix $E^H(\theta) = E$, so $h^{-1} \circ \varphi = \text{id}_E$ is the identity map. That is, $\varphi = h \in H$. So for any $\varphi \in H_{E^H}$ we have $\varphi = h \in H$ so $H_{E^H} \leq H$. Conversely, note that any $h \in H$ fix E^H by definition so $H \leq H_{E^H}$ and combined we have $H = H_{E^H}$. This finishes the proof of property (iii), which implies (i) and (ii).

It remains to show (iv) and (v). To show (iv), we first assume that $H \leq \text{Gal}(E/K)$ is a normal subgroup. Let $\alpha \in E^H$ and let $\beta \in E$ be any conjugate of α over K . By Lemma 1.1 there is some $\phi \in \text{Gal}(E/K)$ such that $\phi(\beta) = \alpha$. Let $h \in H$ be arbitrary, then we find for such h some h' in H such that $\phi \circ h = h' \circ \phi$ (since H is a normal subgroup). It follows that $\phi \circ h(\beta) = h' \circ \phi(\beta) = h'(\alpha) = \alpha$ so after applying ϕ^{-1} on both sides we have $h(\beta) = \phi^{-1}(\alpha) = \beta$. That is, every $h \in H$ fix β so $\beta \in E^H$. Since α was arbitrary in E^H and since β was an arbitrary conjugate of α over K we conclude that E^H is a finite normal extension of K . To see the converse, let $K \leq L \leq E$ be a finite normal extension of K and let H_L be the subgroup of the Galois group that fix L . Any $\phi \in \text{Gal}(E/K)$ will map $\alpha \in L$ to a conjugate of α over K (since ϕ fix K , prove this!), so since L is a finite normal extension of K we have $\phi(\alpha) \in L$ for all $\alpha \in L$. It follows that for any $h \in H_L$ we have $\phi^{-1} \circ h \circ \phi(\alpha) = \phi^{-1}(h(\phi(\alpha))) = \phi^{-1}(\phi(\alpha)) = \alpha$ where we have used that $\phi(\alpha) \in L$ so $h(\phi(\alpha)) = \phi(\alpha)$. We conclude that for any $h \in H_L$ the map $\phi^{-1} \circ h \circ \phi$ fix every element of L , so $\phi^{-1} \circ h \circ \phi \in H_L$ which implies that H is a normal subgroup.

Finally, if $K \leq L \leq E$ is a finite normal extension then we need to show that $\text{Gal}(L/K) \cong \text{Gal}(E/K)/H_L$. To see this, note that (as above) if $\phi \in \text{Gal}(E/K)$ then $\phi(L) = L$ since L is a finite normal extension of K (ϕ is going to map any $\gamma \in L$ to a conjugate of γ over K which is contained in L since L is a finite normal extension). It follows that we can define a homomorphism of groups $\Psi : \text{Gal}(E/K) \rightarrow \text{Gal}(L/K)$ by restriction: $\Psi(\phi) = \phi|_L$. We begin by noting that the kernel of this map is precisely H_L since $\Psi(\phi) = \text{id}_L$ if and only if ϕ fix every element of L . Part (v) now follows

from the fundamental homomorphism theorem (for groups) if we can show that Ψ is a surjective map. That Ψ is surjective follows from the extension theorem. Indeed, let $\psi \in \text{Gal}(L/K)$ then by the extension theorem we can extend ψ to some map $\tilde{\psi} : E \rightarrow E'$ such that $\tilde{\psi}|_L = \psi$. In particular this implies that $\tilde{\psi}(a) = a$ for all $a \in K$ since $K \leq L$ and ψ fix K . Now we claim that $E' = E$. This is proved precisely as in the proof of Lemma 1.1: If $\gamma \in E$ then $\tilde{\psi}(\gamma)$ is a conjugate of γ (since $\tilde{\psi}$ fix K), and since E is a finite normal extension it follows that $\tilde{\psi}(\gamma) \in E$. So, we have $\tilde{\psi} : E \rightarrow E$ and $\tilde{\psi}|_K = \text{id}_K$ so $\tilde{\psi} \in \text{Gal}(E/K)$. Finally, note that $\Psi(\tilde{\psi}) = \tilde{\psi}|_L = \psi$ which proves that $\psi \in \text{Gal}(L/K)$ is in the image of Ψ so Ψ is surjective. ■

1.2. Algebraic numbers produced as radicals and solvable Galois groups.

We will now give a condition for an algebraic number being expressed as radicals. We will prove that the splitting field (for the corresponding irreducible polynomial) for such an algebraic number is always a solvable group, and next class we will produce a polynomial whose splitting field is not solvable. We begin by defining what it means for an algebraic number to be solvable in radicals, and then we start the technical work of proving that the splitting field of the irreducible polynomial of such a number is always a solvable group.

Definition 1.1. Let $\alpha \in \overline{\mathbb{Q}}$ be any algebraic number. We say that α is solvable by radicals if there are numbers $\alpha_1, \dots, \alpha_m \in \overline{\mathbb{Q}}$ and integers $n_1, \dots, n_m$ such that $\alpha_1^{n_1} \in \mathbb{Q}$, for $j < n$ we have $\alpha_{j+1}^{n_{j+1}} \in \mathbb{Q}(\alpha_1, \dots, \alpha_j)$, and $\alpha \in \mathbb{Q}(\alpha_1, \dots, \alpha_n)$.

Example 1.1. The idea in the definition of α being solvable by radicals is that α can be obtained by algebraic operations (taking sums and products), as well as taking roots. To see an example of this, let $p, q \in \mathbb{Q}$ be rational and consider the quadratic polynomial:

$$(1.4) \quad f(x) = x^2 + px + q.$$

By the quadratic formula we can find the zeros of $f(x)$ as:

$$(1.5) \quad \alpha = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q}.$$

So, in this case we can define $\alpha_1 = \sqrt{p^2/4 - q}$ and then have $\alpha_1^2 \in \mathbb{Q}$. We then see that $\alpha \in \mathbb{Q}(\alpha_1)$ so with $\alpha_1 = \sqrt{p^2/4 - q}$ and $n_1 = 2$ we see that α is solvable by radicals. That is, any zero of a degree 2 polynomial is solvable by radicals. This turns out to be true for zeros of degree 3 polynomials and degree 4 polynomials as well, but we will prove that it is not, in general, true for degree 5 polynomials (it is in general not true for polynomials of degree $n \geq 5$). Moreover, notice that the reason that we could prove that the zeros of $f(x)$ were solvable by radicals is that we have a solution formula for degree 2 polynomials (the quadratic formula). So, when we show that zeros of degree 5 polynomials are not in general solvable by radicals we also prove that there can not exist a solution formula for general 5th degree polynomials (or, at least not in radicals).

We now start the somewhat technical work of proving that the Galois group of the irreducible polynomial $f_\alpha(x)$ for some α that is solvable in radicals is always a solvable group.

Lemma 1.3. *Let $E/\mathbb{Q}$ be a finite normal extension and let:*

$$(1.6) \quad U_{n,E} = \{\omega \in E \mid \omega^n = 1\}$$

be all n th roots of unity contained in E . Then:

$$(1.7) \quad p(x) = \prod_{\omega \in U_{n,E}} (x - \omega)$$

is a polynomial with rational coefficients.

Proof. We claim that for any $\omega \in U_{n,E}$ and any $\psi \in \text{Gal}(E/\mathbb{Q})$ we have $\psi(\omega) \in U_{n,E}$. That is, any element of the Galois group maps $U_{n,E}$ to $U_{n,E}$. To see this, let $\omega \in U_{n,E}$ be arbitrary and let $f_\omega(x) \in \mathbb{Q}[x]$ be the irreducible polynomial for ω . Notice that ω is a zero of the polynomial $x^n - 1$, from which it follows that $f_\omega(x)$ divides $x^n - 1$. In particular, any zero of $f_\omega(x)$ is also a zero for $x^n - 1$. It follows that any zero of $f_\omega(x)$ is a root of unity (note that the solutions of $x^n - 1 = 0$ are precisely the n th roots of unity, and $f_\omega(x)$ divides $x^n - 1$). If $\psi \in \text{Gal}(E/\mathbb{Q})$ then $\psi(\omega)$ is also a zero for $f_\omega(x)$ (since ψ fix $\mathbb{Q}$ and $f_\omega(x)$ has rational coefficients). It follows that $\psi(\omega) \in E$ is a n th root of unity so $\psi(\omega) \in U_{n,E}$.

Let $\psi \in \text{Gal}(E/\mathbb{Q})$ be arbitrary. We recall that we have an isomorphism $\psi : E[x] \rightarrow E[x]$ defined by $\psi(a_0 + a_1x + \dots + a_nx^n) = \psi(a_0) + \psi(a_1)x + \dots + \psi(a_n)x^n$. We can apply this to $p(x)$ and obtain:

$$\begin{aligned} \psi(p(x)) &= \psi \left(\prod_{\omega \in U_{n,E}} (x - \omega) \right) = \prod_{\omega \in U_{n,E}} (x - \psi(\omega)) = \prod_{\omega \in \psi^{-1}(U_{n,E})} (x - \omega) = \\ &\quad \prod_{\omega \in U_{n,E}} (x - \omega) = p(x) \end{aligned}$$

where we have used that $\psi^{-1}(U_{n,E}) = U_{n,E}$ since ψ maps any element of $U_{n,E}$ to an element of $U_{n,E}$. It follows that if $p(x) = a_0 + a_1x + \dots + a_mx^m$ then for any $\psi \in \text{Gal}(E/\mathbb{Q})$:

$$(1.8) \quad \psi(a_0) + \psi(a_1)x + \dots + \psi(a_m)x^m = \psi(p(x)) = p(x) = a_0 + a_1x + \dots + a_mx^m,$$

and after comparing the coefficients on both sides of the equality we have $\psi(a_j) = a_j$ for all $j = 1, \dots, m$. Since $\psi \in \text{Gal}(E/\mathbb{Q})$ was arbitrary we conclude that each coefficient of $p(x)$ is fixed by every element of the Galois group. By the fundamental theorem of Galois theory we know that $\mathbb{Q} = E^{H_{\mathbb{Q}}}$ where $H_{\mathbb{Q}}$ is the subgroup of the Galois group that fix every element of $\mathbb{Q}$, and $E^{H_{\mathbb{Q}}}$ is the subfield of E that is fixed by $H_{\mathbb{Q}}$. Since every element of $\text{Gal}(E/\mathbb{Q})$ fix $\mathbb{Q}$, by definition, it follows that $H_{\mathbb{Q}} = \text{Gal}(E/\mathbb{Q})$. We conclude that the subfield that is fixed by $\text{Gal}(E/\mathbb{Q})$ is precisely $\mathbb{Q}$. But since each coefficient of $p(x)$ is fixed by all of $\text{Gal}(E/\mathbb{Q})$ we conclude that $p(x)$ has coefficients in $\mathbb{Q}$. ■

Next we need two lemmas that allow us to show that certain normal extensions have abelian Galois group.

Lemma 1.4. *Let $\mathbb{Q} \leq K \leq E$ be extensions where both K and E are finite normal extensions of $\mathbb{Q}$. If $U_{n,E} = \{\omega_1, \dots, \omega_m\}$ are the n th roots of unity in E then $K(\omega_1, \dots, \omega_m) \leq E$ is a finite normal extension of $\mathbb{Q}$ and K . Moreover, the Galois group $\text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ is an abelian group.*

Proof. We begin by noting that K is a finite extension of $\mathbb{Q}$, so by the primitive element theorem we find some $\theta \in K$ such that $\mathbb{Q}(\theta) = K$. If we let $f_\theta(x) \in \mathbb{Q}[x]$ be the irreducible polynomial for θ then $K = E_{f_\theta}$ is the splitting field of $f_\theta(x)$ (since $\mathbb{Q}(\theta) \leq E_{f_\theta}$ but $K = \mathbb{Q}(\theta)$ is already a finite normal extension so $f_\theta(x)$ splits in $\mathbb{Q}(\theta)$). By Lemma 1.3 we know that $p(x) = (x - \omega_1) \dots (x - \omega_m) \in \mathbb{Q}[x]$. It follows that $f_\theta(x)p(x) \in \mathbb{Q}[x]$. The final observation now is that $K(\omega_1, \dots, \omega_m)$ is the splitting field of $f_\theta(x)p(x)$ which implies that $K(\omega_1, \dots, \omega_m)$ is a finite normal extension of $\mathbb{Q}$. Moreover, we see that $K(\omega_1, \dots, \omega_m)$ is the splitting field of $p(x)$ over K , so $K(\omega_1, \dots, \omega_m)$ is a finite normal extension of K .

It remains to show that $\text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ is abelian. Note that $U_{n,E} = \{\omega_1, \dots, \omega_m\}$ is a group under multiplication: if $\omega_i, \omega_j \in U_{n,E}$ then $\omega_i \omega_j \in E$ (since E is closed under multiplication) and $\omega_i \omega_j$ is still a n th root of unity so $\omega_i \omega_j \in U_{n,E}$. Moreover, we have $1 \in E$ so $1 \in U_{n,E}$ and if $\omega_i \in U_{n,E}$ then $1/\omega_i \in E$ and $1/\omega_i$ is still a n th root of unity so $1/\omega_i \in U_{n,E}$. That is $U_{n,E} \leq U_n$ is a subgroup of the n th roots of unity. As such, since U_n is cyclic, we now that $U_{n,E}$ is a cyclic group generated by some $\omega_1 \in U_{n,E}$ (assume that we have ordered the elements so that ω_1 is a generator). As we say in the proof of the previous lemma, for any $\psi \in \text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ we have $\psi : U_{n,E} \rightarrow U_{n,E}$ and since ψ respects multiplication we have $\psi(\omega_i \omega_j) = \psi(\omega_i) \psi(\omega_j)$ so each $\psi \in \text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ defines a group homomorphism of $U_{n,E}$. That is, we get a map $\text{Gal}(K(\omega_1, \dots, \omega_m)/K) \rightarrow \text{Aut}(U_{n,E})$ where $\text{Aut}(U_{n,E})$ are all isomorphisms (of groups) from $U_{n,E}$ to $U_{n,E}$. This map is a homomorphism since the group operation in both $\text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ and $\text{Aut}(U_{n,E})$ is composition. So we have a homomorphism

$$(1.9) \quad \Phi : \text{Gal}(K(\omega_1, \dots, \omega_m)/K) \rightarrow \text{Aut}(U_{n,E})$$

defined by $\Phi(\psi) = \psi|_{U_{n,E}} : U_{n,E} \rightarrow U_{n,E}$. We claim that Φ is injective. Indeed, if $\Phi(\psi) = \text{id}_{U_{n,E}}$ then $\psi(\omega_j) = \omega_j$ for every $\omega_j \in U_{n,E}$ but then ψ fix K and each $\omega_1, \dots, \omega_m$ so ψ also fix $K(\omega_1, \dots, \omega_m)$, but then ψ is just the identity map on $K(\omega_1, \dots, \omega_m)$. Finally, we claim that $\text{Aut}(U_{n,E})$ is an abelian group which finally implies that $\text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ is abelian since its isomorphic to a subgroup of an abelian group. To see that $\text{Aut}(U_{n,E})$ is abelian we first note that $U_{n,E} \cong \mathbb{Z}_r$ for some integer $r \geq 1$ since $U_{n,E}$ is cyclic. The automorphism group of $\mathbb{Z}_r$ (when we view $\mathbb{Z}_r$ as a group!) is precisely $\mathbb{Z}_r^\times$ where $\mathbb{Z}_r^\times$ are the units in $\mathbb{Z}_r$. Any element $a \in \mathbb{Z}_r^\times$ defines an automorphism by $x \mapsto ax$ which is invertible since a has a multiplicative inverse in $\mathbb{Z}_r$. If $\psi : \mathbb{Z}_r \rightarrow \mathbb{Z}_r$ is any automorphism of the group $\mathbb{Z}_r$ then $\psi(1) = a$ must be a generator of $\mathbb{Z}_r$, and since a is a generator of $\mathbb{Z}_r$ we must have $a \in \mathbb{Z}_r^\times$ (why? Prove this, if a was not a unit then there is some $n < r$ such that $an = 0$). We conclude

that $\psi(1) = a \cdot 1$ so ψ is equal to multiplication by a at 1, but since 1 generate all of $\mathbb{Z}_r$, this implies that $\psi(x) = ax$ for any $x \in \mathbb{Z}_r$.

Finally we can finish the proof. We have an injective homomorphism of groups $\Phi : \text{Gal}(K(\omega_1, \dots, \omega_m)/K) \rightarrow \text{Aut}(U_{n,E})$ and since $\text{Aut}(U_{n,E}) \cong \mathbb{Z}_r^\times$ for some $r \geq 1$ we know that $\text{Aut}(U_{n,E})$ is abelian. It follows that $\text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ is isomorphic as a group to some subgroup of an abelian group, and $\text{Gal}(K(\omega_1, \dots, \omega_m)/K)$ is therefore abelian. $\blacksquare$

Lemma 1.5. *Let $\mathbb{Q} \leq K$ be a finite normal extension of $\mathbb{Q}$. Let $\alpha \in \overline{\mathbb{Q}}$ be an algebraic number such that $\alpha^n \in K$, and let $f_\alpha(x)$ be the irreducible polynomial for α over $\mathbb{Q}$. Let $\alpha_1, \dots, \alpha_m$ be the zeros of $f_\alpha(x)$ and let $E = K(\alpha_1, \dots, \alpha_m)$. If K contains all the n th roots of unity in E then E is a finite normal extension of $\mathbb{Q}$ and K , and $\text{Gal}(E/K)$ is abelian.*

Proof. To prove that E is a finite normal extension of $\mathbb{Q}$ we note that we can write $K = \mathbb{Q}(\theta)$ for some $\theta \in K$. If $f_\theta(x) \in \mathbb{Q}[x]$ is the irreducible polynomial for θ , then we see that E is the splitting field for $f_\theta(x)f_\alpha(x)$, so E is a finite normal extension of $\mathbb{Q}$ since its the splitting field of a polynomial with coefficients in $\mathbb{Q}$. To see that E is a finite normal extension of K , note that E is the splitting field of $f_\alpha(x) \in \mathbb{Q}[x] \subset K[x]$ over K .

It remains to prove that $\text{Gal}(E/K)$ is abelian. Consider the polynomial $x^n - \alpha^n \in K[x]$, since $\alpha^n \in K$ by assumption. Let $\beta_1, \dots, \beta_N \in E$ be the zeros of $x^n - \alpha^n$ that are contained in E . We claim that:

$$(1.10) \quad \{\beta_1, \dots, \beta_N\} = \{\omega\alpha \mid \omega \in U_{n,E}\}.$$

One direction of this is clear: if $\omega \in U_{n,E}$ then $\omega \in E$ and $(\omega\alpha)^n - \alpha^n = \omega^n\alpha^n - \alpha^n = 1 \cdot \alpha^n - \alpha^n = 0$ so $\omega\alpha$ is a zero for $x^n - \alpha^n$. To get the converse, assume that $\beta_j \in E$ is any zero of $x^n - \alpha^n$. Since the zeros of $x^n - \alpha^n$ are precisely:

$$\alpha, e^{2\pi i/n}\alpha, e^{2\pi i \cdot 2/n}\alpha, \dots, e^{2\pi i(n-1)/n}\alpha$$

we conclude that there must be some n th root of unity ω such that $\beta_j = \omega\alpha$. However, since $\alpha, \beta_j \in E$ we have $\omega = \beta_j/\alpha \in E$ so $\omega \in U_{n,E}$. That is, if β_j is a zero of $x^n - \alpha^n$ that is contained in E then $\beta_j = \omega\alpha$ for some $\omega \in U_{n,E}$.

We have done the calculation for the polynomial $x^n - \alpha^n$, however if α_j is any conjugate of α over $\mathbb{Q}$ (i.e., any zero of $f_\alpha(x)$), then we claim that $\alpha_j^n \in K$ as well, which shows that the calculation done above works equally well for any conjugate α_j of α . Indeed, since K is a finite normal extension of $\mathbb{Q}$ we know that if $\gamma \in K$ then for any $\psi \in \text{Gal}(E/\mathbb{Q})$ we know that $\psi(\gamma)$ is a conjugate of γ (since ψ fix $\mathbb{Q}$, so it fix the irreducible polynomial for γ , $f_\gamma(x) \in \mathbb{Q}[x]$). However, if K is a normal extension then any conjugate of $\gamma \in K$ is contained in K so $\psi(\gamma) \in K$. That is, $\psi(K) = K$ (note that ψ may not fix the elements in K , but it maps the set into itself). Now, for any conjugate of α_j of α there is some $\psi \in \text{Gal}(E/\mathbb{Q})$ such that $\psi(\alpha) = \alpha_j$ and $\alpha^n \in K$ so:

$$K \ni \psi(\alpha^n) = \psi(\alpha)^n = \alpha_j^n.$$

Now, if $\psi \in \text{Gal}(E/K)$ then ψ maps the zeros of $x^n - \alpha_j^n \in K[x]$ to zeros of $x^n - \alpha_j^n$ since ψ fix K and $x^n - \alpha_j^n$ has coefficients in K . It follows that for every $j = 1, \dots, m$ and any $\psi \in \text{Gal}(E/K)$ there is some $\omega_{\psi,j} \in U_{n,E}$ such that $\psi(\alpha_j) = \omega_{\psi,j}\alpha_j$. Moreover, since $U_{n,E} \subset K$ by assumption and since $\psi, \varphi \in \text{Gal}(E/K)$ fix K we have:

$$(1.11) \quad \omega_{\psi \circ \varphi, j} \alpha_j = \psi \circ \varphi(\alpha_j) = \psi(\omega_{\varphi, j} \alpha_j) = \psi(\omega_{\varphi, j}) \psi(\alpha_j) = \omega_{\varphi, j} \omega_{\psi, j} \alpha_j$$

or: $\omega_{\psi \circ \varphi, j} = \omega_{\varphi, j} \omega_{\psi, j}$. In other words, we have proved that the map $q_j : \text{Gal}(E/L) \rightarrow U_{n,E}$ defined by $q_j(\psi) = \omega_{\psi, j}$ is a homomorphism of groups. Let q be the map defined by:

$$(1.12) \quad q : \text{Gal}(E/K) \rightarrow U_{n,E} \times U_{n,E} \times \dots \times U_{n,E}, \quad q(\psi) = (q_1(\psi), \dots, q_m(\psi))$$

where we have taking products of m copies of $U_{n,E}$. Since each q_j is a homomorphism of groups we know that q is a homomorphism of groups. Moreover, if $q(\psi) = (1, \dots, 1)$ then $q_j(\psi) = 1$ for every j , or $\psi(\alpha_j) = \alpha_j$ for every j . That is, if $q(\psi) = (1, \dots, 1)$ then ψ fix K (since $\psi \in \text{Gal}(E/K)$), and ψ fix every α_j so ψ fix every element of $E = K(\alpha_1, \dots, \alpha_m)$. That is, if $q(\psi) = (1, \dots, 1)$ then ψ is the identity map on E , so the kernel of q is trivial and we conclude that q is an injective group homomorphism. Since each group $U_{n,E}$ is abelian we conclude that $U_{n,E} \times \dots \times U_{n,E}$ is abelian, and since q is injective we conclude that $\text{Gal}(E/K)$ is isomorphic to the subgroup of an abelian group and is therefore abelian. $\blacksquare$

With these two technical lemmas we are now ready to prove the following important theorem:

Theorem 1.2. *Let $\alpha \in \overline{\mathbb{Q}}$ be solvable in radicals. If $f_\alpha(x) \in \mathbb{Q}[x]$ is the irreducible polynomial for α , and if E_α is the splitting field of $f_\alpha(x)$, then $\text{Gal}(E_\alpha/\mathbb{Q})$ is a solvable group.*

Proof. Since α is solvable in radicals we find $\alpha_1, \dots, \alpha_m \in \overline{\mathbb{Q}}$ and integers $n_1, \dots, n_m$ such that $\alpha_{j+1}^{n_{j+1}} \in \mathbb{Q}(\alpha_1, \dots, \alpha_j)$, for $j < m$, and $\alpha \in \mathbb{Q}(\alpha_1, \dots, \alpha_m)$. For each $j = 1, \dots, m$ we let $f_j(x) \in \mathbb{Q}[x]$ be the irreducible polynomial for α_j . We define a sequence of fields $K_0, K_1, K_2, \dots, K_m$ by letting $K_0 = \mathbb{Q}$ and K_j be the splitting field for:

$$(1.13) \quad f_1(x)f_2(x)\dots f_j(x) \in \mathbb{Q}[x].$$

Note that each K_j is a finite normal extension of $\mathbb{Q}$ since each K_j is a splitting field of a polynomial with coefficients in $\mathbb{Q}$. Moreover, from the definition we obtain $K_j \leq K_{j+1}$ since any zero of $f_1(x)\dots f_j(x)$ is also a zero of $f_1(x)\dots f_j(x)f_{j+1}(x)$. Our first goal is to show that $\text{Gal}(K_m/\mathbb{Q})$ is solvable, which will then use to prove that $\text{Gal}(E_\alpha/\mathbb{Q})$ is solvable.

Let U_{n_j, K_j} be the n_j th roots of unity contained in K_j . We define $p_j(x)$ by:

$$(1.14) \quad p_j(x) = \prod_{\omega \in U_{n_j, K_j}} (x - \omega)$$

which by Lemma 1.3 is a polynomial with rational coefficients, $p_j(x) \in \mathbb{Q}[x]$. We define fields L_j as the splitting field of the polynomial:

$$(1.15) \quad f_1(x)f_2(x)\dots f_j(x)p_j(x) \in \mathbb{Q}[x].$$

Notice that each L_j is a finite normal extension of $\mathbb{Q}$ since it is the splitting field of a rational polynomial, and $K_{j-1} \leq L_j \leq K_j$ since the zeros of $p_j(x)$ are, by assumption, contained in K_j . That is, we have produced a sequence of extensions:

$$(1.16) \quad K_0 \leq L_1 \leq K_1 \leq L_2 \leq K_2 \leq \dots \leq L_n \leq K_n,$$

where every K_j and L_j is a finite normal extension of $\mathbb{Q}$. Moreover, note that each K_j is obtained from L_j by adding the zeros of $f_j(x)$, so K_j is a splitting field over L_j and similarly L_{j+1} is obtained by adding the zeros of $p_{j+1}(x)$ to K_j so L_{j+1} is a splitting field over K_j . That is, K_j/L_j is also a finite normal extension and L_j/K_{j-1} is a finite normal extension.

By the fundamental theorem of Galois theory each K_j and L_j corresponds to a normal subgroup of $\text{Gal}(K_n/\mathbb{Q})$, we write them as $H_{K_j} = G_j$ and $H_{L_j} = H_j$. Moreover, since $L_j \leq K_j$ any element of $\text{Gal}(K_n/\mathbb{Q})$ that fix K_j also fix L_j so we have $G_j \trianglelefteq H_j$ and since $K_j \leq L_{j+1}$ any element that fix L_{j+1} also fix K_j so $H_{j+1} \trianglelefteq G_j$. Moreover, note that $G_n = \{\text{id}_{K_n}\}$ since by definition an element of G_n fix all of K_n and $G_0 = \text{Gal}(K_n/\mathbb{Q})$ since an element of G_0 fix only $K_0 = \mathbb{Q}$, but any element of the Galois group fix all of $\mathbb{Q}$. That is, we have produced a sequence of normal subgroups of $\text{Gal}(K_n/\mathbb{Q})$:

$$\{\text{id}_{K_n}\} = G_n \trianglelefteq H_n \trianglelefteq G_{n-1} \trianglelefteq H_{n-1} \trianglelefteq G_{n-2} \trianglelefteq \dots \trianglelefteq G_1 \trianglelefteq H_1 \trianglelefteq G_0 = \text{Gal}(K_n/\mathbb{Q}).$$

To prove that $\text{Gal}(K_n/\mathbb{Q})$ is solvable it remains to prove that the quotients H_j/G_j and G_{j-1}/H_j are all abelian groups, this then proves that $\text{Gal}(K_n/\mathbb{Q})$ is solvable by definition (since it has a subnormal series with abelian quotients). Note that $H_j = \text{Gal}(K_n/L_j)$ and $G_j = \text{Gal}(K_n/K_j)$ so if we apply the fundamental theorem of Galois theory (property (v)) with $L_j \leq K_j \leq K_n$ then we obtain:

$$(1.17) \quad H_j/G_j = \text{Gal}(K_n/L_j)/\text{Gal}(K_n/K_j) \cong \text{Gal}(K_j/L_j),$$

so we need to prove that $\text{Gal}(K_j/L_j)$ is abelian. Note that $\alpha_j^n \in K_{j-1} \subset L_j$ and K_j is obtained from L_j by adding all conjugates of α_j to L_j . Moreover, L_j is contained from K_{j-1} by adding the zeros of $p_j(x)$ which are precisely the n_j th roots of unity in K_j , so L_j contains all of the n_j th roots of unity in K_j . It follows that the assumptions of Lemma 1.5 are satisfied, and Lemma 1.5 implies that $\text{Gal}(K_j/L_j)$ is abelian. similarly, $G_{j-1}/H_j \cong \text{Gal}(L_j/K_{j-1})$ and since L_j is obtained from K_{j-1} by adding the n_j th roots of unity in K_j so Lemma 1.4 implies that $G_{j-1}/H_j \cong \text{Gal}(L_j/K_{j-1})$ is abelian. This finishes the proof that $\text{Gal}(K_n/\mathbb{Q})$ is a solvable group.

To show that $\text{Gal}(E_\alpha/\mathbb{Q})$ is solvable, note that $E_\alpha \leq K_n$. Indeed, K_n is a finite normal extension of $\mathbb{Q}$ and $\alpha \in \mathbb{Q}(\alpha_1, \dots, \alpha_n) \leq K_n$ so since $f_\alpha(x)$ has zero in K_n it follows that every zero of $f_\alpha(x)$ is within K_n . But this implies that $E_\alpha \leq K_n$ since E_α is generated by the zeros of $f_\alpha(x)$. Since E_α is the splitting field of $f_\alpha(x)$ it follows that $E_\alpha/\mathbb{Q}$ is a finite normal extension, so by the fundamental theorem of Galois theory (part (v)) it holds that:

$$(1.18) \quad \text{Gal}(E_\alpha/\mathbb{Q}) \cong \text{Gal}(K_n/\mathbb{Q})/H_{E_\alpha}.$$

In particular, $\text{Gal}(E_\alpha/\mathbb{Q})$ is isomorphic to a quotient of a solvable group, but this implies that $\text{Gal}(E_\alpha/\mathbb{Q})$ is also solvable (why? Try to prove this, it follows from the characterization of solvable groups using the derived series). $\blacksquare$

By Theorem 1.2 we know that if $\alpha \in \overline{\mathbb{Q}}$ is solvable in radicals, then $\text{Gal}(E_\alpha/\mathbb{Q})$ is a solvable group. Next class we will find an explicit polynomial $f(x) \in \mathbb{Q}[x]$ of degree 5 such that the splitting field of $f(x)$ has a Galois group isomorphic to S_5 . Since S_5 is not a solvable group, this implies that the zeros of $f(x)$ are not solvable in radicals, and in extension it shows that we can not have a solution formula for the zeros of a polynomial of degree 5 in radicals (that is, there can not exist a generalization of the quadratic formula to polynomials of degree 5).

Email address: sandfeldt@uchicago.edu